INCH - POUND

MIL-C-47002A(MI)
24 May 1991
SUPERSEDING
MIL-C-47002(MI)
19 April 1974

MILITARY SPECIFICATION

COATING, POLYURETHANE, SINGLE COMPONENT SYSTEM

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a polyurethane, single component coating and encapsulation system compound.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standard, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5970

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SPECIFICATIONS

FEDERAL

PPP-P-1892	-	Paint, Varnish, Lacquer, and Related Materials, Packaging, Packing, and Marking of

STANDARDS

FEDERAL

FED-STD-313	-	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 115	-	Varnishes Used for Electrical Insulation, Standard Methods of Testing
ASTM D 150	-	A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials, Standard Test Methods for
ASTM D 257	-	D-C Resistance or Conductance of Insulating Materials, Standard Test Methods for
ASTM D 3273	-	Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber, Standard Test Method for

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material. The coating compound shall be a single component, air curing, polyurethane formulation capable of being fully cured within 720 hours (30 days).

3.3 Electrical properties. The compound shall meet the requirements specified in table I at 25 ± 3°Celsius (C) (77 ± 5.4°Fahrenheit [F]) when tested in accordance with 4.7.1.

TABLE I. Electrical properties.

Property	Values	Applicable Test
Dielectric strength (volts/mil)	Not less than 2500	4.7.1.1
Dielectric constant (60 to 10^6 Hertz [Hz])	Not greater than 3.6	4.7.1.2
Dissipation factor (60 to 10^6 Hz)	Not greater than 0.005	4.7.1.3
Volume resistivity (10^{13} ohm-cm)	Not less than 4	4.7.1.4
Surface resistivity (megohm per square)	Not less than 7500	4.7.1.5

3.4 Physical properties. The material shall meet the requirements specified in 3.4.1 through 3.4.5 when tested in accordance with 4.7.2.

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3.4.1 Appearance and color. The compound shall be homogeneous and free of lumps and coarse particles. A skin is permissible but shall be removed and discarded before use. The compound shall be yellowish in color (see 4.7.2.1).

3.4.2 Fungus resistance. The material shall show no evidence of fungus growth (see 4.7.2.2).

3.4.3 Shelf life. The shelf life of the material in unopened containers shall be not less than 6 months from date of manufacture when stored at temperature less than 27°C (80.6°F) (see 4.7.2.3).

3.4.4 Moisture resistance. The moisture resistance shall be not less than 100,000 megohms (see 4.7.2.4).

3.4.5 Viscosity. The viscosity of the material shall be in the range of 4,000 to 4,300 millipascal seconds (mPa·s) (see 4.7.2.5).

3.5 Workmanship. The workmanship shall be such as to insure a product which is uniform, in conformance with this specification, and properly packaged and packed. The compound shall be free of dirt, foreign material, or other contamination. Containers shall be clean, uniformly filled, well sealed, and legibly marked.

3.6 Material safety data sheets. A material safety data sheet shall be prepared in accordance with the requirements of FED-STD-313 (see 6.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of

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known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in ASTM D 115, ASTM D 150, ASTM D 257, ASTM D 3273 and MIL-STD-202 and as specified herein.

4.4 First article inspection. When specified in the contract or purchase order (see 6.2) a first article sample of a quantity of compound sufficient to conduct all tests specified herein shall be provided. The first article sample shall be fabricated using the same processes, facilities, and materials as are proposed for production. The first article shall meet all requirements of this specification. No failures shall be permitted during first article sample inspection. If a failure occurs the first article sample shall be rejected. No units shall be offered for acceptance until approval of a first article sample has been obtained from the procuring activity.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table I.

TABLE I. Quality conformance inspection.

Quality	Requirement paragraph	Test paragraph
1. Appearance and color	3.4.1	4.7.2.1
2. Viscosity	3.4.5	4.7.2.5

4.6 Lot and sampling.

4.6.1 Lot. A lot shall consist of all the material submitted for acceptance at the same time, which has been prepared by the same company without change in materials or processes in one continuous period of operation.

4.6.2 Sampling. Sampling shall be in accordance with the contract or order (see 6.2). Any failure of a sample shall cause the entire lot to be rejected.

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4.7 Test methods.

4.7.1 Electrical property tests.

4.7.1.1 Dielectric strength. The dielectric strength shall be determined in accordance with ASTM D 115 for compliance with 3.3.

4.7.1.2 Dielectric constant. The dielectric constant shall be determined in accordance with ASTM D 150 for compliance with 3.3.

4.7.1.3 Dissipation factor. The dissipation factor shall be determined in accordance with ASTM D 150 for compliance with 3.3.

4.7.1.4 Volume resistivity. The volume resistivity shall be determined in accordance with ASTM D 257 for compliance with 3.3.

4.7.1.5 Surface resistivity. The surface resistivity shall be determined in accordance with ASTM D 257 for compliance with 3.3.

4.7.2 Physical property tests.

4.7.2.1 Appearance and color. The appearance and color shall be determined visually in accordance with 3.4.1.

4.7.2.2 Fungus resistance. The material shall be tested for fungus resistance in accordance with ASTM D 3273.

4.7.2.3 Shelf life. The compound shall be tested for conformance with all the requirements specified herein after completion of storage period.

4.7.2.4 Moisture resistance. The moisture resistance shall be determined in accordance with MIL-STD-202, Method 106, after completing ten 24 hours cycles under polarizing voltages.

4.7.2.5 Viscosity. The viscosity of the material shall be determined in accordance with ASTM D 115 using a Brookfield viscometer, Model RVF (or equivalent), with a Number 3 spindle at 15 revolutions per minute.

4.8 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with PPP-P-1892.

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5. PACKAGING

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be in accordance with PPP-P-1892, as specified by the procurement document (see 6.2).

5.2 Marking. Each container of material shall be legibly and durably marked in accordance with MIL-STD-129 (see 6.2). Marking information shall include, but shall not necessarily be limited to, the following:

a. Manufacturers product designation

b. Manufacturers name and address

c. Lot or batch number

d. Date of manufacture

e. Shelf life

f. Necessary precautionary notices

g. Cure cycle.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The compound covered by this specification is intended for use in coating circuit boards by dipping, brushing or spraying. The compound, in fully cured condition, is effective in temperature ranges of -4°C to + 121°C.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. Whether a first article sample is required (see 3.1 and 4.4)

d. Sampling plan (see 4.6.2)

e. Applicable level and requirements for preservation, packaging, and packing (see 5.1)

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f. Marking requirements, if other than specified (see 5.2).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.7. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 Subject term (keyword) listing.

Encapsulation material
Insulation, electrical
Varnish, electrical component

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Reviewing activity:
DLA - GS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-47002A (MI)	910514

3. DOCUMENT TITLE
COATING, POLYURETHANE, SINGLE COMPONENT SYSTEM

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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